

Getting Started with Imaclim-S

A Beginner's Guide to Documenting and Running Imaclim-S Model

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1 Introduction

1.1 What is Imaclim?

Imaclim is a macroeconomic model designed to analyze the interactions between the economy, energy systems, and climate policies. It is widely used for:

- Evaluating the economic impacts of climate policies (e.g., carbon taxes)
- Simulating energy transitions
- Producing long-term economic projections under different scenarios

Imaclim is typically implemented in **Scilab**, an open-source numerical computing environment.

1.2 Prerequisites

Before you start, make sure you have:

1. **Scilab** installed — <https://www.scilab.org>
2. **Python** installed (for post-processing results and producing charts)
3. The **IMACLIM-Country** repository cloned locally from GitHub: https://github.com/CIRED/IMACLIM-Country/tree/tutorial_imaclim

Important: This tutorial is based on the branch **tutorial_imaclim**. Make sure you cloned on this branch before proceeding.

1.3 Key Documentation

Two essential references for understanding how the model works:

- **Gaëlle Le Treut's thesis** (most up-to-date summary of IMACLIM-S): <https://hal.science/tel-01707559>
 - **IMACLIM equations reference document** (equations of the IMACLIM-S model): [IMACLIM-S equations](#)
-

2 Repository Structure

Once cloned, the IMACLIM-Country repository is organised as follows:

```
IMACLIM-Country/

code/                                # Scilab scripts to run the model
  ImaclimS.sce                       # Run this to launch an IMACLIM-S simulation
  outputs_full_template.sce         # Saves output variables
  Test_AllConfig/                   # Scripts to run multiple variants
    run_dashboards.sce              # Creates dashboards with variants
    run_modelisations.sce           # Runs all variant simulations
    def_of_MacroIncFRA.sce          # Declares variants to test
    SCNBC_RunChoice3.sce            # Contains variant logic for scenarios

study_frames/                        # All model input variables
  study_frames_FRA2018/
    Dashboard_FRA2018.csv           # Main dashboard (entry point for variables)
    Country_Selection.csv           # Select which country to study
    AMSrun3mixnote/                 # Scenario-specific input matrices
      IOT_Qtities_AMSrun3mixnote_1.csv # I/O table for 2030
      IOT_Qtities_AMSrun3mixnote_2.csv # I/O table for 2040
      IOT_Qtities_AMSrun3mixnote_3.csv # I/O table for 2050

Library/
  economic_equations.sci           # All model equations

outputs/                             # Results written here after each run
  AMSrun3mixnote.../
    FullTemplate_AMSrun3mixnote.csv # All variables, all years
    Dashboard_FRA2018.csv           # Dashboard used for this run
```

The repository is organised around four main folders:

- **code/** contains all the Scilab scripts that drive the model — **ImaclimS.sce** is the main entry point for a single simulation, while the **Test_AllConfig/** subfolder contains the scripts needed to run multiple variants in batch.
- **study_frames/** holds all the input data for the model: the Dashboard sets the scenario configuration, **Country_Selection.csv** defines the country studied, and the scenario subfolders (e.g. **AMSrun3mixnote/**) contain the input-output tables for each projection year (2030, 2040, 2050).
- **Library/** contains the core economic equations of IMACLIM-S — useful to consult when interpreting model behaviour.
- **outputs/** is where all simulation results are written after each run, with one subfolder per scenario containing the full results template and the dashboard used for that run.

3 Understanding the Imaclim Model Structure

3.1 Key Concepts

Before running scenarios, you should understand the three main components of Imaclim:

1. Calibration The model is calibrated to a **base year** (2018 for France). This means all model parameters are set so that the model exactly reproduces observed economic data for that year. You can skip re-calibration on subsequent runs by setting `skip_calibration;True` in the dashboard — this saves significant computation time.

2. Exogenous Variables These are variables that are set *outside* the model — typically projections about the future that you feed in. Examples include:

- Population growth and labour productivity
- Household saving rates
- Energy and world commodity prices
- Carbon tax settings

Note: If a variable is not present in the resolution system (in the folder `code/Systeme_Resolution`), it is treated as exogenous by default.

3. Scenarios A scenario defines a particular set of assumptions and policy choices. The main scenario used in this tutorial is `AMStrun3mixnote`. You can find the input tables of this scenario in the folder `study_frames/study_frames_FRA2018/AMStrun3mixnote/`. Variants are created by changing specific variables (e.g., household consumption in the aviation sector).

3.2 The Dashboard

The **Dashboard** (`study_frames/study_frames_FRA2018/Dashboard_FRA2018.csv`) is the central control file for any simulation. It contains all the variables that drive the model. Here is an annotated excerpt of the most important variables:

```
/// Simulation ///
```

<code>Scenario;AMStrun3mixnote</code>	<code>// Name of the scenario</code>
<code>Nb_Iter;3</code>	<code>// Number of iterations (2018 to 2050)</code>
<code>skip_calibration;True</code>	<code>// Skip calibration to save time</code>

```
/// Households ///
```

<code>H_DISAGG;HH1</code>	<code>// Number of household income groups (1-10 deciles)</code>
<code>VAR_saving;ref</code>	<code>// Variant on the household saving rate</code>

```
/// Sector aggregation ///
```

<code>AGG_type;AGG_23TME</code>	<code>// Use the 23-sector version of the model</code>
---------------------------------	--

```
/// Carbon tax ///
```

<code>Carbone_ETS;False</code>	<code>// No ETS carbon pricing</code>
<code>ClosCarbRev;CstNetLend</code>	<code>// Carbon revenue recycling option</code>

```

/// Productivity ///
Capital_Dynamics;%F          // No capital stock dynamics
Demographic_shift;True       // Productivity depends on demographics
Labour_product;True          // Productivity depends on active population

/// Energy & commodity prices ///
World_prices;True            // Import energy prices active (gas, oil...)

/// Variant simulations ///
study;SNBC3_RunChoices3      // RunChoice file used for variants

```

3.3 The Reference Scenario

The reference scenario (**ref**) is always your baseline. All other scenarios are **compared against it** to measure the impact of a policy.

4 Running Your First Simulation

4.1 Step 1 — Select the Country

If this is your first time using the model, open the file `study_frames/Country_Selection.csv` and uncomment the country you want to study (here FRA2018):

```

// For a study on France:
Country;France;FRA2018

```

4.2 Step 2 — Launch the Simulation

Open Scilab, navigate to the `code` folder, and run the main simulation file:

```

cd('IMACLIM-Country\code')
clear
exec ImaclimS.sce

```

This launches a simulation using the dashboard at: `study_frames/study_frames_FRA2018/Dashboard_FRA2018.csv`. The command `clear` ensures you start with a clean workspace.

The model will run through the calibration phase (if not skipped) and then produce projections for each time step (2030, 2040, 2050). Depending on your computer's processing power, this may take several minutes.

Results are saved automatically to the `outputs/` folder.

4.3 Step 3 — Explore the Results

Once the simulation finishes, a new directory with all the results will be created in `outputs/` — e.g. `outputs/AMRun3mixnote/`. The key output files to explore are:

File
<code>outputs/AMRun3mixnote/FullTemplate_AMRun3mixnote.csv</code>
<code>outputs/AMRun3mixnote/Time_2030/ioQ-run_2030_AMRun3mixnote.csv</code>

The output folder is composed of three main files:

- `Dashboard_FRA2018.csv`: the dashboard used for this run (copy of the one in `study_frames`)
- `FullTemplate_AMRun3mixnote.csv`: the full template with all variables for all years
- Folders prediction for each time step (2030, 2040, 2050) with key variables for that year (e.g. `ioQ-run_2030_AMRun3mixnote.csv`)

For each time step (here 2030, 2040, 2050), you have the prediction for each key variable of the model:

- **ioQ**: input-output table quantities
- **Prices**: prices of each good
- **ecoT**: main variables consumption of investment, consumption, netlending of each agents (households, firms, government) of the model
- **CO2Emiss**: CO2 emissions by sector
- **TechCoeff**: technical coefficients of the model (capital productivity, labour productivity...)
- **Trad_sectors**: trade flows by sector

Note: If a file contains `run` in its name, it corresponds to the model's projection for the year studied. If the file contains `ini`, this corresponds to the result of the previous time step that was used to compute the projection of the current one. IMACLIM-S is a recursive model, so the projection for 2040 depends on the results of 2030, and the projection for 2050 depends on the results of 2040.

5 Analysing an Economic Shock

One of Imacsim's core strengths is simulating the macroeconomic impact of a **shock** — a deliberate change to one or more input variables. This section walks through a concrete example: **analysing the macroeconomic impact of a -10% and -20% reduction in household consumption in the aviation sector**.

The aviation sector corresponds to row 19 of the household consumption vector: `Proj_Vol.C.val(19)`.

Step 1 — Declare the variants in the RunChoice file

Open code/Test_AllConfig/SCNBC_RunChoice3.sce and add the variant logic for your new variable:

```
// Modify aviation household consumption (row 19 = aviation sector)
if Scenario == "AMStrun3mixnote"
    if VAR_C_airtransp == "ref"          // Reference case
        Proj_Vol.C.val(19) = Proj_Vol.C.val(19);

    elseif VAR_C_airtransp == "-10%"    // -10% aviation consumption
        Proj_Vol.C.val(19) = Proj_Vol.C.val(19) * 0.9;

    elseif VAR_C_airtransp == "-20%"    // -20% aviation consumption
        Proj_Vol.C.val(19) = Proj_Vol.C.val(19) * 0.8;
    end
end
```

Step 2 — Register the variants to test in def_of_MacroIncFRA.sce

Open code/Test_AllConfig/def_of_MacroIncFRA.sce and add your variable:

```
test_fra_macro.VAR_C_airtransp = ['ref', '-10%', '-20%'];
```

Note: The file def_of_MacroIncFRA.sce is used to create the different variants of the scenario. This file is used to create the different dashboards in the step 6.

Step 3 — Add the variable to the default dashboard class

Open code/Test_AllConfig/Class_default_dashboard.sci and declare the variable:

```
default_dashboard = struct( ..
    'Carbone_ETS' , 'False', ..
    ...
    'VAR_C_airtransp', '');
```

Step 4 — Add the new variable to the main Dashboard CSV

Open study_frames/study_frames_FRA2018/Dashboard_FRA2018.csv and add:

VAR_C_airtransp;-20%

This step is important to follow the trace of your variable in the dashboard used for the simulation.

Step 5 — Add the new variable to the output template (optional)

If you want to track your new variable in the output template, you can add it in the file `code/outputs_full_template.sce`:

```
["VAR_C_airtransp", VAR_C_airtransp];
```

Step 6 — Generate the dashboards

```
cd('IMACLIM-Country\code\Test_AllConfig')  
exec run_dashboards.sce
```

This creates an intermediate file `code/Test_AllConfig/dashboards.dat` containing one dashboard per variant. You can inspect them in Scilab:

```
dashboards      // lists all dashboards  
dashboards(1)   // shows the first dashboard - check your changes are there
```

Step 7 — Run all variant simulations

```
exec run_simulations.sce
```

This runs each dashboard in turn and saves results for each variant in the `outputs/` folder. You can then study and compare the results of the simulations in the output files.

6 Resources

- Scilab documentation: <https://www.scilab.org/documentation>
- IMACLIM-Country GitHub repository: <https://github.com/GaelleLeTreut/IMACLIM-Country>
- Gaëlle Le Treut's thesis (IMACLIM-S reference): <https://hal.science/tel-01707559>
- IMACLIM equations document: https://www.centre-cired.fr/wp-content/uploads/2020/09/cired_wp_2020_85_le_treut.pdf